

Lecture 1 for the seminar on

Matroids and the integral Hodge conjecture for abelian varieties

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1 The picture

The paper proves a failure of the integral Hodge conjecture for very general principally polarized abelian varieties of dimension at least 4, and also for intermediate Jacobians of very general cubic threefolds. The key is that the obstruction is detected by *monodromy* of certain degenerations, and that this monodromy is encoded by *regular matroids*.

2 Hodge conjecture for abelian surfaces

2.1 Minimal classes

Let (X, Θ) be a principally polarized abelian variety of dimension g . The *minimal class* for curves is

$$\frac{\Theta^{g-1}}{(g-1)!} \in H^{2g-2}(A, \mathbb{Z}).$$

For a very general X , the Hodge classes of curve type are generated by this class, so the integral Hodge conjecture becomes the question:

Which integral multiples of the minimal class are represented by algebraic curves?

The paper shows that, in the very general cases considered, *only even multiples can occur*.

2.2 Main theorem I

The first statement is the following.

Theorem (Theorem 1.1). *Let X be a very general PPAV of dimension $g \geq 4$. Then every algebraic cycle Z of codimension $g - 1$ has cohomology class*

$$[Z] = m \cdot \frac{[\Theta]^{g-1}}{(g-1)!} \in H^{2g-2}(X, \mathbb{Z})$$

with m even.

So the integral cycle class map does not hit the primitive generator. In particular, the integral Hodge conjecture fails for these abelian varieties. In fact, Theorem 1.1 holds for cycles of codimension $2 \leq c \leq g - 1$. For example, if we have proved Theorem 1.1 for $g = 4$ and $c = 3$, then the case $c = 2$ follows from the simple observation that

$$\Theta \cdot \frac{\Theta^2}{2} = 3 \frac{\Theta^3}{3!}$$

is an odd multiple of the minimal curve class on A . Then $[Z \cdot \Theta] = 3m \cdot \frac{[\Theta]^3}{3!}$ and $3m$ is even.

2.3 Main theorem II

The same method for proving Theorem 1.1 applies to intermediate Jacobians of cubic threefolds.

Theorem (Theorem 1.3). *Let Y be a very general smooth cubic threefold, and let*

$$J(Y) = H^3(Y, \mathbb{C}) / (F^2 H^3(Y) \oplus H^3(X, \mathbb{Z})_{tf})$$

be its intermediate Jacobian. Then every curve C on $J(Y)$ has homology class

$$[C] = m \cdot \frac{[\Theta_Y]^4}{4!} \in H_2(J(Y), \mathbb{Z})$$

with m even.

Before explaining the impact of this theorem, we first recall the story of Clemens-Griffiths. It is classically known by Clemens-Griffiths that the rationality problem for rationally connected threefolds Y can be solved by studying the intermediate Jacobian $J(Y)$ of Y , which is a PPAV.

For cubic threefolds, Clemens-Griffiths proves that $J(Y)$ is not a product of Jacobians of curves, which is enough to prove that Y is not rational. This is due to another theorem of Clemens-Griffiths stating that if there exists a birational mapping between Y and $\mathbb{P}^3$, then $J(Y) \cong J(C)$ for some (possibly reducible) non-singular curve C .

Back to our story, the minimal class is involved by Matsusaka's criterion:

The minimal class $\frac{\Theta^{g-1}}{(g-1)!}$ is represented by an effective algebraic curve
 $\iff$ an indecomposable PPAV is a Jacobian of curves.

So the result of Clemens-Griffiths implies that the minimal class is not represented by an effective curve. Here the theorem proves something stronger for a *very general* cubic threefold: it is not represented by any algebraic 1-cycle at all.

This stronger statement matters because it connects to stable rationality, which is discovered by Voisin.

Theorem (Voisin). *Let Y be a rationally connected threefold. Then Y admits a (cohomological) decomposition of the diagonal iff:*

1. $\text{Tors}(H^3(Y, \mathbb{Z})) = 0$ ¹.
2. *There exists a universal codimension 2 cycle on $J(Y) \times Y$.*
3. *There exists a 1-cycle $Z \in J(Y)$ such that $[Z] = \frac{[\Theta]^{g-1}}{(g-1)!}$, where $g = \dim J(Y)$.*

The existence of such decompositions is a necessary criterion for stable rationality.

Theorem (Corollary 1.4). *A very general cubic threefold does not admit a decomposition of the diagonal. Hence it is not stably rational.*

2.4 Embedding into Jacobians

A further corollary of the main theorems involves another method for probing the Hodge conjecture. To be explicit, Voisin proves that

An abelian variety X is a direct summand in a product of Jacobians $\iff$
 $X \times \widehat{X}$ satisfies the integral Hodge conjecture for curve classes.

Beckmann et. al. proves that the integral Hodge conjecture holds for products of Jacobians of curves, and hence for their direct summands. Hence it is not surprising to have the following corollary.

Theorem (Corollary 1.5). *Let (X, Θ) be one of the following varieties:*

1. *a very general PPAV;*
2. *the intermediate Jacobian of a very general cubic threefold.*

Then for any abelian variety Y , any isogeny $f: X \times Y \rightarrow \prod J(C)$ to a product of Jacobians of curves has even degree. In particular, X does not admit a split embedding into a product of Jacobians.

¹Artin-Mumford stably rationality criterion

3 Why matroids enter the story

3.1 Slogan

The authors study specially chosen degenerations of PPAVs over a punctured poly-disk. In such degenerations, the monodromy around each coordinate hyperplane gives a quadratic form on a lattice, and the data of a collection of linear forms whose squares describe these monodromy forms are packaged exactly by a regular matroid.

Very roughly:

1. a regular matroid M gives a list of integral linear forms;
2. from those linear forms one builds a *matroidal family* of abelian varieties;
3. if a very general fiber contains a curve with odd minimal-class multiplicity, then monodromy would force a certain "quadratic splitting" of M into data coming from a graph;
4. the combinatorial theorem says this can happen only when M is cographic.

This is the conceptual slogan: algebraicity of a curve class forces a decomposition property, and that decomposition property has a purely combinatorial obstruction.

3.2 Main combinatorial theorem

The combinatorial theorem we need is the following.

Theorem (Theorem 1.6). *Let M be a regular matroid of rank g , and let $\mathcal{X} \rightarrow \Delta$ be a matroidal family of g -dimensional PPAVs associated to M . Let $f: C_t \rightarrow X_t$ be a projective curve in a very general fiber with*

$$f_*[C_t] = m \cdot \frac{[\Theta]^{g-1}}{(g-1)!} \in H_2(X_t, \mathbb{Z}).$$

If M is not cographic, then m must be even.

So the existence of an odd multiple of the minimal class is obstructed by non-cographicity.

Sketch of Proof of Theorem 1.1. We first assume that $g = 4$. Consider the graphic matroid $M(K_5)$, which is not planar, hence is not cographic. K_5 has 5 vertices and 10 edges, thus the rank of $M(K_5)$ is

$$\text{rk} M(K_5) = |\text{edge}(M(K_5))| - 1 = 4.$$

Hence $M(K_5)$ corresponds to families of abelian fourfolds as the dimension g of the PPAV family is the rank of the matroid. Now apply Theorem 1.6.

For higher dimension, make induction by degenerating X to $Y \times E$ where E is an elliptic curve. $\square$

Sketch of Proof of Theorem 1.3. Similar to Theorem 1.1, consider the Segre cubic threefold

$$Y_0 := \left\{ \sum_{i=0}^5 x_i = \sum_{i=0}^5 x_i^3 = 0 \right\} \subset \mathbb{P}^5,$$

which is the unique cubic threefold with 10 nodes. Construct a flat family $Y \rightarrow B$ with central fiber Y_0 , such that the restriction $Y_{\Delta^{10}} \rightarrow \Delta^{10}$ is a universal deformation of Y_0 . Let $J(Y^*) \rightarrow B^*$ be the associated family of intermediate Jacobians. A result of Gwena says that it is associated to the matroid R_{10} , which is not cographic. $\square$

3.3 Review: weight filtration

We now recall some basic definitions and results introduced last semester. Given a degeneration family $f: \mathcal{X} \rightarrow \Delta$, with generic fiber an abelian g -fold, consider the VHS

$$V = H_1(X_t, \mathbb{Z}).$$

By Schmid's nilpotent orbit theorem, the degeneration has a limit MHS, and the weight filtration is induced by the log monodromy operator N ,² which satisfies the condition

$$N^2 = 0.$$

In other words, the filtration can only have a "width" of 2. Since the "pure" weight of the first homology H_1 is -1 , we conclude that the range of the weight filtration is $\{-2, -1, 0\}$. In general, the graded pieces have geometric meanings:

1. gr_0^W is the discrete part: 1-cycles that are dual to the vanishing cycles. These roughly correspond to the component group of the singular fiber.
2. gr_{-1}^W is the abelian part: 1-cycles that has compact support on the singular fiber. This is the H_1 of the abelian quotient of the semi-abelian limit, i.e. invariants under the monodromy.
3. gr_{-2}^W is the toric part (vanishing cycles): 1-cycles that are loops around the nodes. They shrink to zero (vanish) in the singular limit. These are the image of N .

²We can take log as locally monodromy operators are always unipotent by general Hodge theory.

As a sidestory, the filtration mirrors the exact sequence of the semi-abelian group scheme G (the Raynaud extension):

$$0 \longrightarrow T \longrightarrow G \longrightarrow B \longrightarrow 0.$$

1. Weight -2 part corresponds to the torus T . Since $H_1(\mathbb{G}_m) \cong \mathbb{Z}(1)$, it carries a Tate twist, shifting the weight from 0 to -2 .
2. Weight -1 part corresponds to the abelian variety B . $H_1(B)$ has the standard weight -1 .
3. Weight 0 part corresponds to the discrete lattice Λ . It appears in Raynaud uniformization, used to recover the original abelian variety X with bad reduction from G in the analytic sense: $X \cong G/\Lambda$. Since it is discrete (copies of points), it has weight 0.

3.4 Review: Mumford construction

The Mumford construction can be roughly stated as follows. The subgroup $W_{-1}V \subset \pi_1(X_t)$ gives rise to a cover $\mathcal{Y}^* \rightarrow \mathcal{X}^*$ fiberwise corresponding to the intermediate cover of the universal cover:

$$\mathbb{C}^g \cong \widetilde{X}_t \longrightarrow Y_t \longrightarrow X_t,$$

and the Galois group of this cover is gr_0^W . Y_t is a semi-abelian variety, which can be regarded as a torus bundle according to the Raynaud extension.

In the case of a maximal degeneration (W_{-1} has rank g), the intermediate cover $\mathcal{Y}^*$ of the punctured family admits an analytic open embedding $\mathcal{Y}^* \hookrightarrow (\mathbb{C}^*)^g \times \mathbb{C}^*$ into an algebraic torus, with the map to Δ^* given by the projection to the second factor. Thus, by the theory of toric varieties, it is possible to extend $\mathcal{Y}^* \rightarrow \Delta^*$ to a family $\mathcal{Y} \rightarrow \Delta$.

3.5 Review: cographic matroid (see also §2.5.2)

The most important example of matroids (ground set + realization preserving independence) is the cographic matroid. The toy model arise from a graph G and let $E = E(G)$ be the set of edges. Choose an orientation on the edges. We have an inclusion

$$H_1(G, \mathbb{Z}) \subset \mathbb{Z}^E$$

as every homology class can be viewed as a $\mathbb{Z}$ -linear combination of directed edges. If e_i is the basis vector of $\mathbb{Z}^E$ corresponding to the i -th edge, we restrict its dual basis to the homology group to get a linear functional

$$\mathbf{x}_i \in H_1(G, \mathbb{Z})^\vee \cong H^1(G, \mathbb{Z}).$$

The cographic matroid $M^*(G)$ on the ground set E has realization

$$E \longrightarrow H^1(G, \mathbb{Z}), \quad i \longmapsto \mathbf{x}_i.$$

Such a cographic matroid is often related to the singularities of varieties. For example, let $\pi : C \rightarrow \Delta^k$ be a family of nodal curves, in which

1. C_0 is a nodal projective curve with k nodes;
2. C_t is a smooth fiber nearby C_0 .

Let $E = \{1, \dots, k\}$ be the set of nodes of C_0 and let $\gamma_1, \dots, \gamma_k \in H_1(C_t, \mathbb{Z})$ be the corresponding vanishing cycles, unique up to sign. A choice of sign is equivalent to a choice of orientation of the edges of $G = \Gamma(C_0)$, the dual graph of C_0 . Each γ_i may be viewed as a linear form on $H_1(C_t, \mathbb{Z})$ which vanishes on W_{-1} ³, and therefore descends to a linear form on gr_0^W .

On the other hand, we have the identification⁴

$$\text{gr}_0^W H_1(C_t, \mathbb{Z}) \cong H_1(\Gamma(C_0), \mathbb{Z}).$$

Thus γ_i is identified with the linear form on $H^1(G, \mathbb{Z})$ giving the coordinate of the oriented edge $e_i \in E(G)$.

Therefore, the matroid realized by the k vanishing cycles in C_t is isomorphic to the cographic matroid $M^*(G)$. Conversely, the cographic matroid $M^*(G)$ associated to any graph G arises this way, because we can construct a nodal projective curve C_0 whose dual complex is G .

As a conclusion, cographic matroids are exactly the ones that arise naturally from degenerations of curves. Since Jacobians come from curves, the later monodromy data on the curve side is cographic. Therefore, if the abelian side can be related to a curve through an odd minimal-class curve class, its matroidal monodromy should have to become cographic in a precise sense.

³Since $W_{-2} = \text{im} N$, $W_{-1} = \ker N$, and the intersection pairing has the property $(Nx, y) + (x, Ny) = 0$, we conclude that $(\gamma, \eta \in W_{-1}) = (Na, \eta) = -(a, N\eta) = 0$.

⁴This requires, for example, strict D-semistable.

4 Idea behind Theorem 1.6

4.1 Slogan

odd multiple of minimal class $\implies$ curve in a very general fibre

$\implies$ orthogonal decomposition of cohomology / homology

$\implies$ quadratic splitting in a graph coming from a degeneration of curves

$\implies$ matroid is cographic.

4.2 Decomposition of cohomology groups

We may assume that C_t is smooth projective by replacing C_t by its normalization. Thus we have a morphism $f: J(C_t) \rightarrow X_t$ by composing $C_t \rightarrow X_t$ with the Abel-Jacobi map $x \mapsto [x - x_0]$. Notice that both sides are PPAV, the dual defines a map

$$f^\vee: X_t \cong \widehat{X_t} \rightarrow \widehat{J(C_t)} \cong J(C_t).$$

Lemma (Lemma 2.9). $f \circ f^\vee: X_t \rightarrow X_t$ is multiplication by m .

For simplicity we consider the cohomology with coefficients $\mathbb{R}$ (in fact, we only need a ring where m is invertible). Hence $\frac{1}{m}f^\vee$ splits the map f_* and there is a canonical decomposition ⁵

$$H_1(C_t, \mathbb{R}) \cong H_1(X_t, \mathbb{R}) \oplus \ker(f_*: H_1(C_t, \mathbb{R}) \rightarrow H_1(X_t, \mathbb{R})). \quad (*)$$

Lemma (Lemma 2.10). *The decomposition (*) is orthogonal with respect to Θ_{C_t} . Explicitly, we have*

$$(f_*^\vee H_1(X_t, \mathbb{R}))^{\perp_{\Theta_{C_t}}} = \ker(f_*: H_1(J(C_t), \mathbb{R}) \rightarrow H_1(X_t, \mathbb{R})).$$

The proof of Lemma 2.8 and Lemma 2.10 are postponed and we continue the idea of the proof of Theorem 1.6. Since $t \in \Delta^*$ is very general, we can spread out the 1-cycle C_t over a Zariski open subset after a generically finite cover. For simplicity, we temporarily ignore the cover and assume that the decomposition (*) holds over a Zariski open subset of Δ^* , and hence is respected by all monodromy operators. (If the decomposition spreads out over the base as a decomposition of local systems, then it is automatically π_1 -equivariant, hence monodromy-invariant.)

⁵Here we use the natural identification $H_1(C_t, \mathbb{Z}) \cong H_1(J(C_t), \mathbb{Z})$.

4.3 Monodromy bilinear forms

We now assume that the matroidal family is given by the matroid M with ground set S . Fix $t \in (\Delta^*)^S$. Thus the corresponding matroidal family can be written as $X^* \rightarrow (\Delta^*)^S$. Denote by T_s the monodromy operator about the s -th coordinate hyperplane. We assume that each T_s is unipotent, so that we can talk about log operators $N_s := \log T_s$.

Then $(*)$ becomes a decomposition of limiting MHS under the weight filtration defined by

$$W_{-1} = \cap \ker N_s, \quad W_{-2} = \left\langle \sum \operatorname{im} N_s \right\rangle.$$

Taking gr_0^W we get

$$\operatorname{gr}_0^W H_1(C_t) \cong \operatorname{gr}_0^W H_1(X_t) \oplus \ker(f_* : \operatorname{gr}_0^W H_1(C_t) \rightarrow \operatorname{gr}_0^W H_1(X_t)).$$

We can thus define the *monodromy bilinear form* Q_s on $\operatorname{gr}_0^W H_1(C_t)$ by ⁶

$$Q_s(u, v) := \Theta(N_s u, v).$$

The decomposition $(*)$ is orthogonal with respect to Q_s , so we may restrict Q_s to $\operatorname{gr}_0^W H_1(X_t)$ and denote the restriction by B_s . Since the family is matroidal, these monodromies are given by a multiple of the bilinear forms $(u, v) \mapsto y_s(u)y_s(v)$, which we denote by y_s^2 . ⁷

Let $\Gamma(X_0)$ be the dual complex of the central fiber X_0 . Then there is a canonical isomorphism

$$\operatorname{gr}_0^W H_1(X_t, \mathbb{Z}) \cong H_1(\Gamma(X_0), \mathbb{Z}).$$

This is Proposition 5.10 in the article we read last semester. In short, the monodromy bilinear forms B_s may be viewed as bilinear forms on $H_1(\Gamma(X_0), \mathbb{Z})$ canonically.

4.4 Quadratic splittings

We can abstract the procedure above and package it into a combinatorial framework.

⁶This is similar to regarding the vanishing cycle around the node as a linear form on gr_0^W , as we stated before. Actually the following procedure is the same as the example of nodal curves.

⁷For a degenerating family of PPAVs, N measures the toric part that appears in the limit. In general, N could encode several interacting vanishing directions, giving a higher-rank quadratic form. But on the matroidal locus, each irreducible boundary component contributes only a rank-one piece. This is also why the compactified Torelli map lands naturally in the matroidal locus: a stable curve degenerates by acquiring nodes, and each node contributes a rank-one form coming from the corresponding edge in the dual graph. The Jacobian degeneration therefore has exactly the sort of monodromy data that matroidal cones parametrize.

Definition. Let M and N (with ground set S, E) be regular matroids with integral realizations

$$\begin{aligned} S &\longrightarrow U^*, & s &\longmapsto y_s, \\ E &\longrightarrow V^*, & e &\longmapsto x_e. \end{aligned}$$

Let Λ be a ring and d be a positive integer. A *quadratic Λ -splitting of level d* of M in N consists of the following data:

1. an embedding $U_\Lambda \hookrightarrow V_\Lambda$;
2. a decomposition $V_\Lambda = U_\Lambda \oplus U'$;
3. a matrix $(a_{se}) \in \mathbb{Z}_{\geq 0}^{S \times E}$ of non-negative integers,

such that for all $s \in S$, the bilinear form $Q_s := \sum_e a_{se} x_e^2$ has the following properties:

1. the decomposition $V_\Lambda = U_\Lambda \oplus U'$ is orthogonal with respect to Q_s ;
2. the restriction of Q_s to U_Λ agrees with $d \cdot y_s^2$.

Hence to prove Theorem 1.6, it suffices to first make reduction to combinatorics (Theorem 1.8), and prove the resulting combinatorial statement (Theorem 1.9). It is worth noticing that Theorem 1.9 is purely combinatorial.

Theorem (Theorem 1.8). *Let $X^* \rightarrow (\Delta^*)^S$ be a matroidal family of PPAVs associated to a regular matroid M as above. Let $f: C_t \rightarrow X_t$ be a projective curve in a very general fiber with*

$$f_*[C_t] = m \cdot \frac{[\Theta]^{g-1}}{(g-1)!} \in H_2(X_t, \mathbb{Z}).$$

Let ℓ be a prime which is coprime to m . Then there is a positive integer d such that M admits a quadratic $\mathbb{Z}_\ell$ -splitting of level d in a cographic matroid.

Theorem (Theorem 1.9). *A regular matroid admits a quadratic $\mathbb{Z}_{(2)}$ -splitting of some level $d \geq 1$ into a cographic matroid if and only if it is itself cographic.*

Proof of Theorem 1.6. If there is a curve on a very general fiber with m odd, then m is invertible in $\mathbb{Z}_{(2)}$. Theorem 1.8 and Theorem 1.9 together imply that M itself is cographic. $\square$

5 Postponed proofs

5.1 Proof of Lemma 2.9

We first claim that $(f^\vee)^*\Theta_C = m \cdot \Theta$. This follows from the Poincaré duality, by which we have the isomorphism

$$H^2(X, \mathbb{R}(1)) \cong H^{2g-2}(\widehat{X}, \mathbb{R}(g-1)) \cong H^{2g-2}(X, \mathbb{R}(g-1)),$$

mapping Θ to $\frac{\Theta^{g-1}}{(g-1)!}$. In other words, a divisor class in H^2 corresponds to a 1-cycle class in H^{2g-2} , and specifically the polarization class Θ corresponds to the minimal class. Now the class $\Theta_C \in H^2(J(C), \mathbb{R}(1))$ corresponds, via Poincaré duality on $J(C)$, to the class of the Abel–Jacobi embedded curve

$$[C] \in H^{2g_c-2}(J(C), \mathbb{R}(g_c-1)).$$

Then functoriality of Poincaré duality implies that $(f^\vee)^*\Theta_C = m \cdot \Theta$ is equivalent to

$$f_*[C] = m \cdot \frac{\Theta^{g-1}}{(g-1)!}.$$

Hence if we denote by ϕ_C and ϕ_X be the principal polarizations on $J(C)$ and X , we have

$$(f^\vee)^*(\phi_C) = m \cdot \phi_X.$$

Hence the composition $f \circ f^\vee$ is multiplication by m . □

5.2 Proof of Lemma 2.10

Lemma 2.9 implies that $\frac{1}{m}f_*^\vee$ splits the map f_* :

$$H_1(C, \mathbb{R}) \cong f_*^\vee H_1(X, \mathbb{R}) \oplus \ker(f_*: H_1(C, \mathbb{R}) \rightarrow H_1(X, \mathbb{R})).$$

Let $\beta \in H_1(X, \mathbb{Z})$. The associated element of $H^1(\widehat{X}, \mathbb{Z})$ is given by $\Theta(\beta, -)$, and its image via $f_*^\vee$ is given by the linear form $L := \Theta(\beta, f_*(-))$.

It is clear that any $\alpha \in \ker(f_*)$ lies in the kernel of L . On the other hand, the restriction of Θ_C to the subspace $f_*^\vee H_1(X, \mathbb{R})$ has to be non-degenerate and unimodular, since m is invertible in $\mathbb{R}$. Thus $\ker L$ cannot be larger than $\ker(f_*)$. □